

b) Loop Unrolling:-

Reduce loop overhead by executing multiple iterations in one loop cycle.

Benefit:-

- Reduces loop control operations.
- Faster execution.

Eg:- Before:-

```
for (int i = 0; i < 4; i++) {
    sum += a[i];
}
```

After Optimization:-

```
sum += a[0];
sum += a[1];
sum += a[2];
sum += a[3];
```

c) Strength Reduction:-

Replace expensive operations with cheaper ones.

Benefit:-

- Multiplication $\rightarrow$ Addition.
- Improves speed.

Eg:- Before:-

```
for (i = 0; i < n; i++) {
    x = i * 2;
}
```

After Optimization:-

```
x = 0;
for (i = 0; i < n; i++) {
    x = x + 2;
}
```